

AUTONOMUS AGENTS AND MULTI-AGENT SYSTEMS COORDINATION

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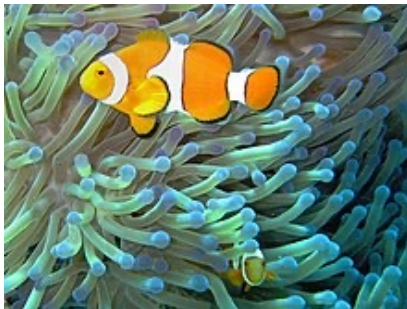
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COLLABORATION VS. COOPERATION

Cooperation

From latin *cooperatus* (to work with)

- Not a defined shared goal
- No shared ownership for the full operation
- Only potentially / partially planned



Collaboration

From latin *co-* (together) and *-laborare* (to work)

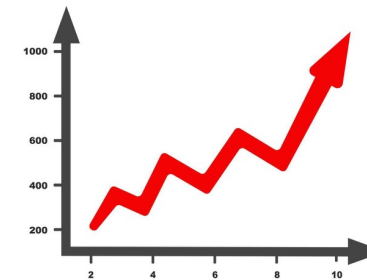
- Shared goal
- Shared ownership
- Planned
- Potentially central control/leader



DECISION MAKING IN GROUPS

- Leader election
- Voting
- Auctions
- Bargaining

Usually requires
preferences or valuations



BULLY ALGORITHM

3 types of messages:

- **Election Message:** Sent to announce election.
- **Answer (Alive) Message:** Responds to the Election message.
- **Coordinator (Victory) Message:** Sent by winner of the election to announce victory.

BULLY ALGORITHM

1. If Agent *A* receives **Victory / Coordinator** message, treat sender as coordinator
2. If Agent *A* has highest ID, broadcast **Victory / Coordinator** message to all others (e.g., when recovering from failure or when coordinator fails)
3. If **Election** message received
 - From higher ID: do nothing → wait for victory
 - From lower ID: send **Answer** and initiate **Election**
4. No response to Election: claim **Victory**

PROBLEMS WITH BULLY

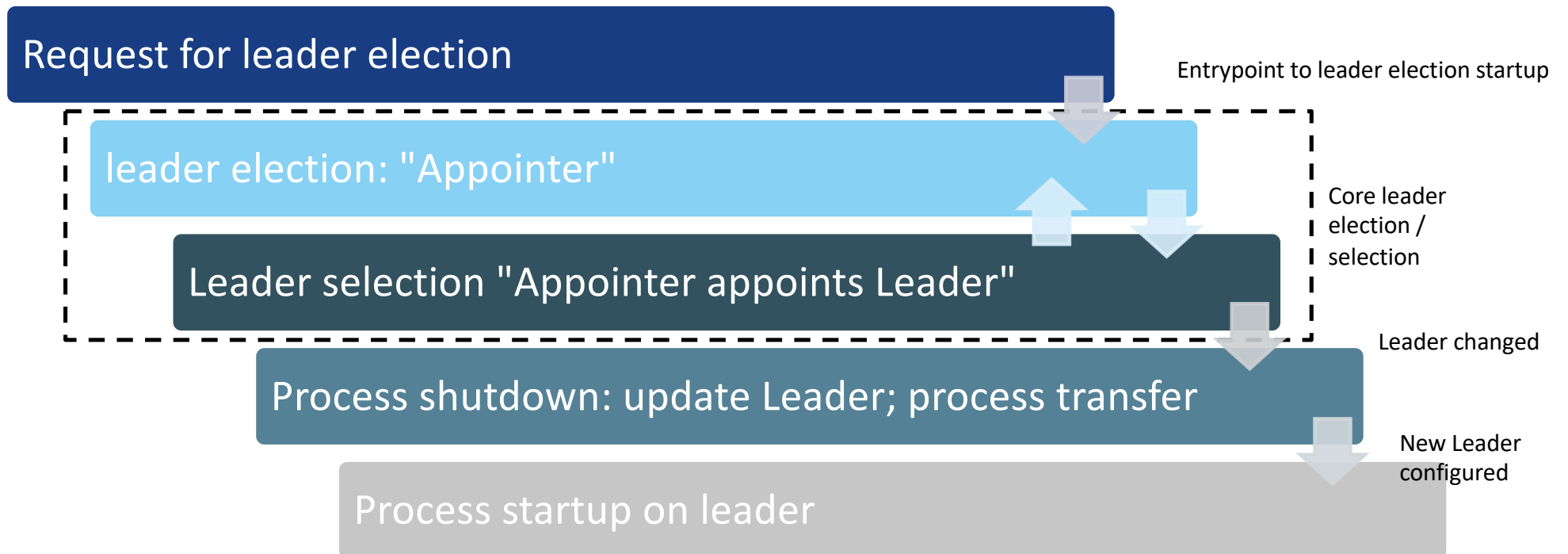
1. All Agents need to be honest
2. New agents might have higher ID but never become leader
3. ID decides leader

LEADER ELECTION

1. Leader election runs continuously in background
2. A unique agent is selected as “Appointer”
3. Appointer queries all other agents for health metrics (convergecast)
4. Appointer appoints leader and informs all agents (broadcast)
5. Process repeats periodically as a failsafe (e.g., leader has died)

How to select appointer??

Rationale: decouple (i) finding a unique agent from (ii) finding the best leader, making leader selection a centralized problem through the appointer.



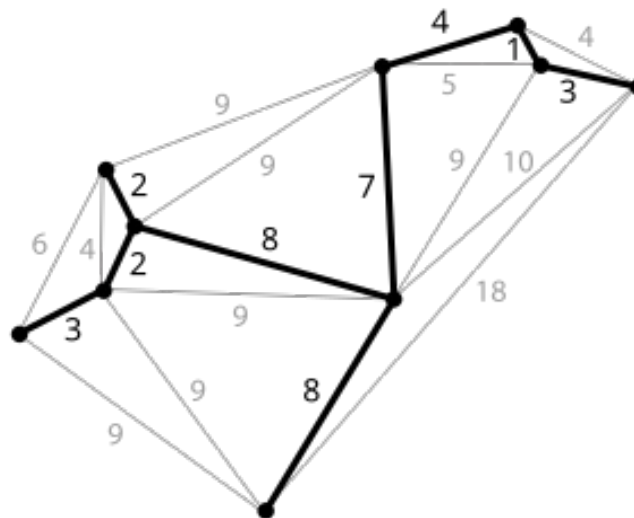
LEADER ELECTION

Distributed Minimum Spanning Tree using Gallager, Humblet, and Spira's algorithm (also known as Yo-Yo algorithm)

0. Everyone is root of its own one node subtree
1. Ping neighbours to find edge cost, send to root along own subtree's MST
2. Nearest* neighbour subtrees merge
 - *defined by edge cost
3. Repeat
 - up to $\log(n)$ times

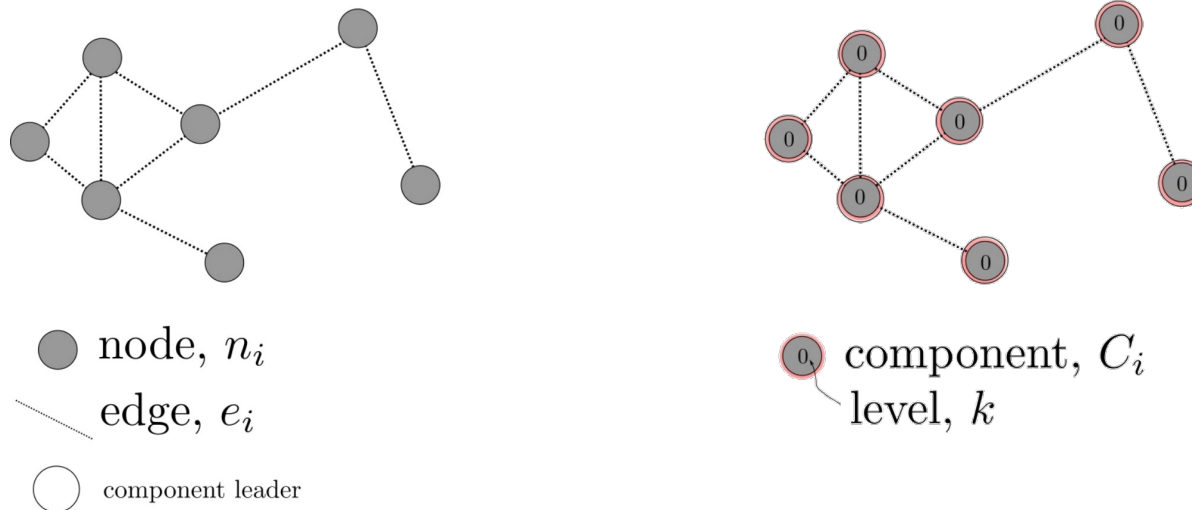
R. G. Gallager, P. A. Humblet, and P. M. Spira, "A distributed algorithm for minimum-weight spanning trees," ACM Trans. Program. Lang. Syst., vol. 5, no. 1, pp. 66–77, Jan. 1983, doi: 10.1145/357195.357200.

MST EXAMPLE



SELECTING APPOINTER

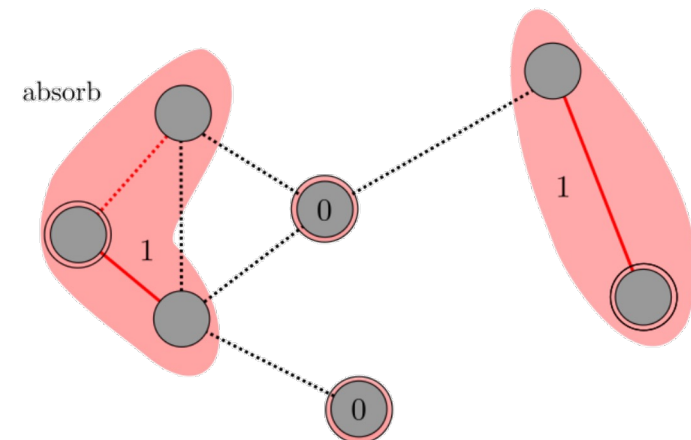
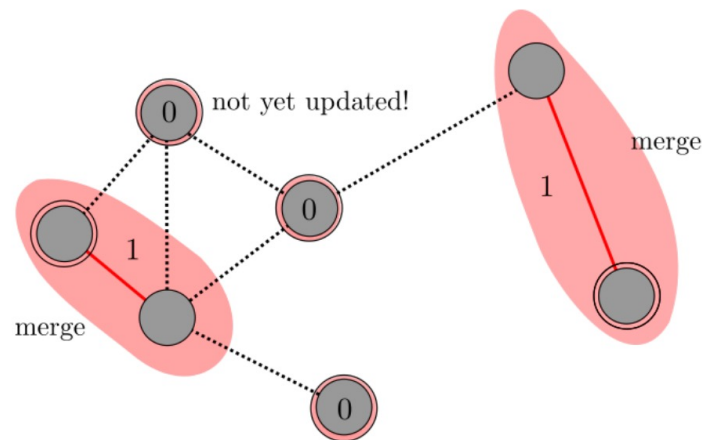
Start with a forest of single agents; initialize each to a level; set leader of this component



SELECTING APPOINTER

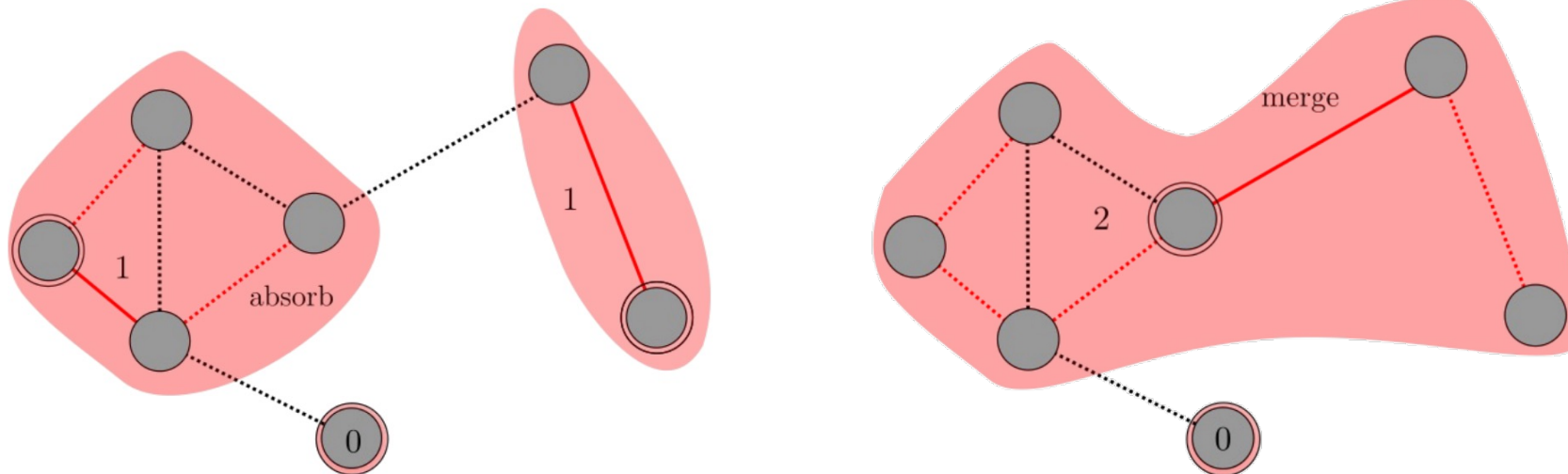
Build up using repeated merge and absorb operations

- Very important: operations can be asynchronous



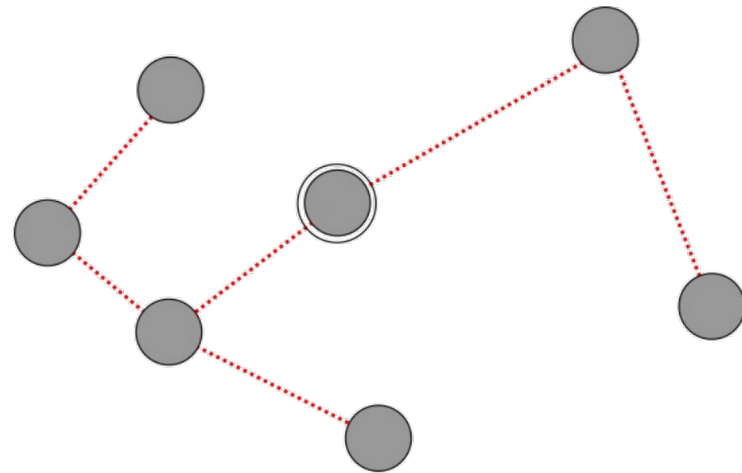
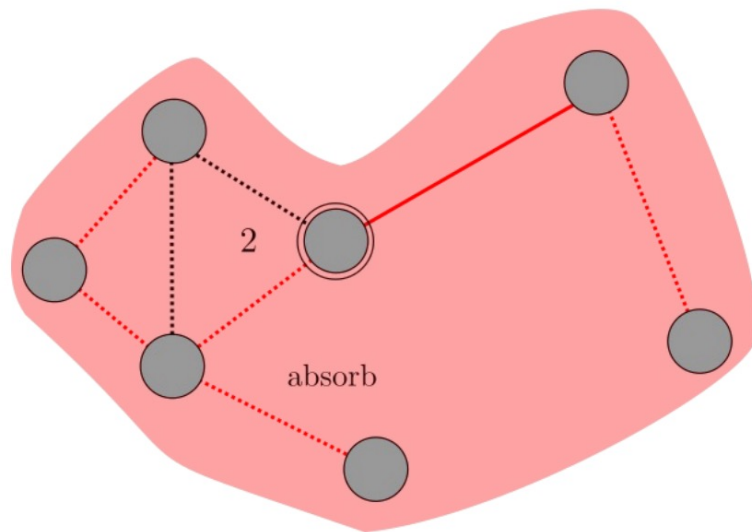
SELECTING APPOINTER

Continue until no new components to probe



SELECTING APPOINTER

Yo-yo output: a rooted MST, which can be used for leader election using a unique “Appointer” node



SELECTING LEADER

Appointer requests health information from all agents

Broadcast request on MST

Wait for convergecast via MST

Appointer decides best leader

Appointer informs all agents of new leader via MST broadcast

VOTING

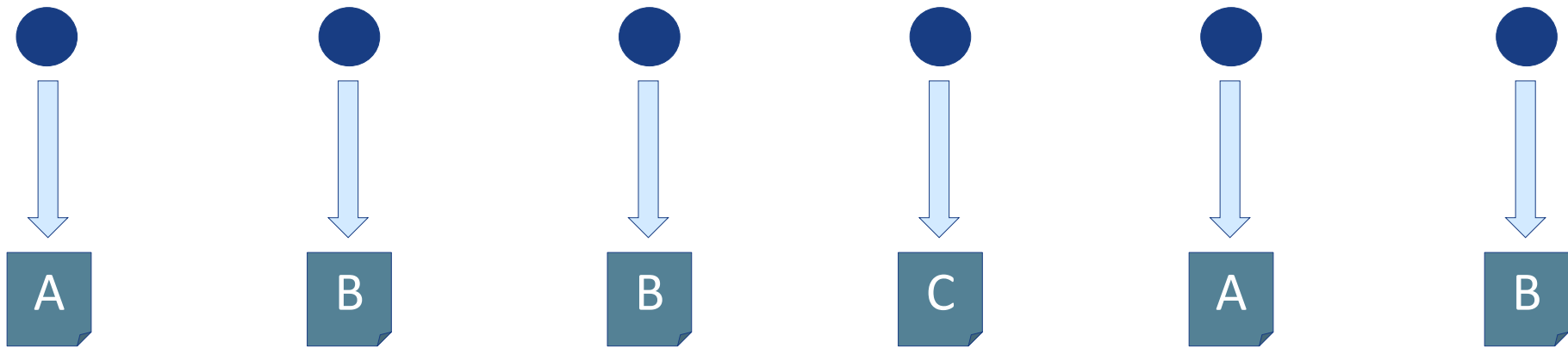
Making a decision as a group based on individual preferences

- Plurality voting
- Sequential majority elections
- Borda count
- Slater ranking

PLURALITY VOTING

Also known as *first-past-the-post*, *single-choice voting*, or *winner-takes-all*

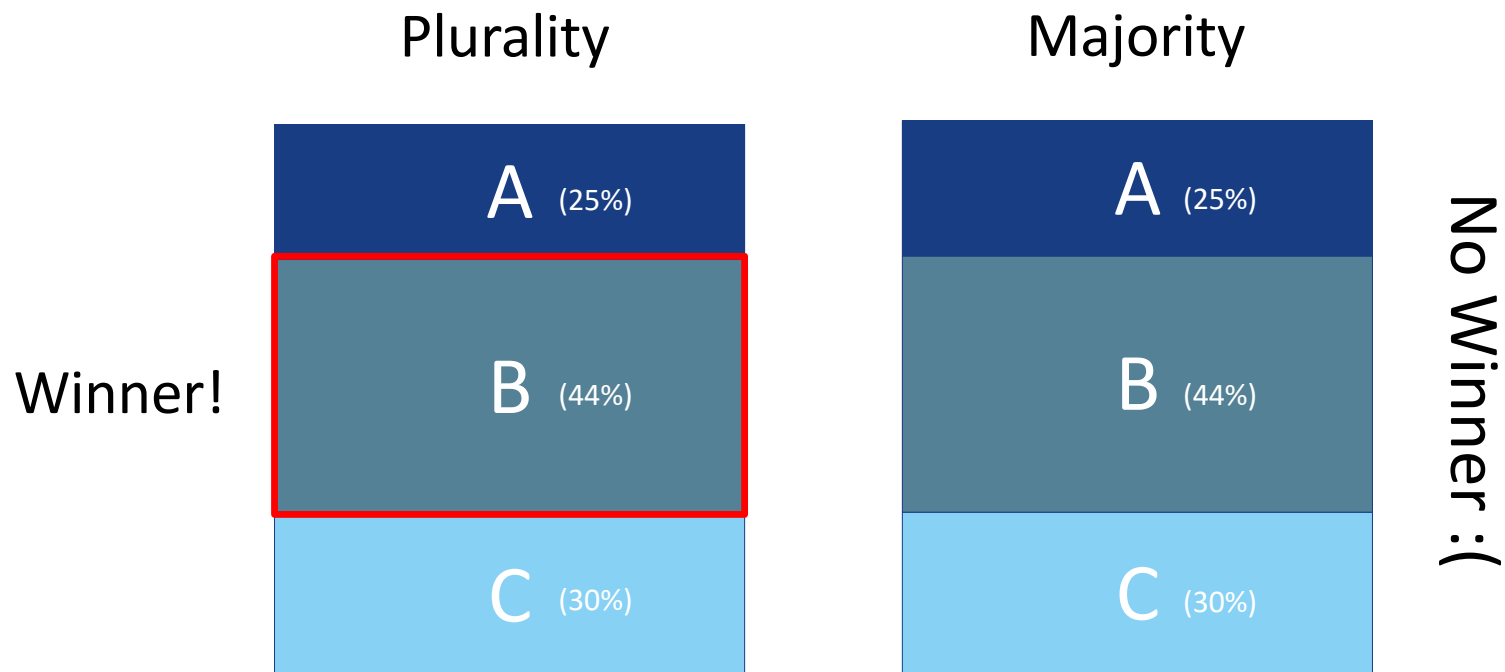
Each agent needs to have a single preference from the set of options



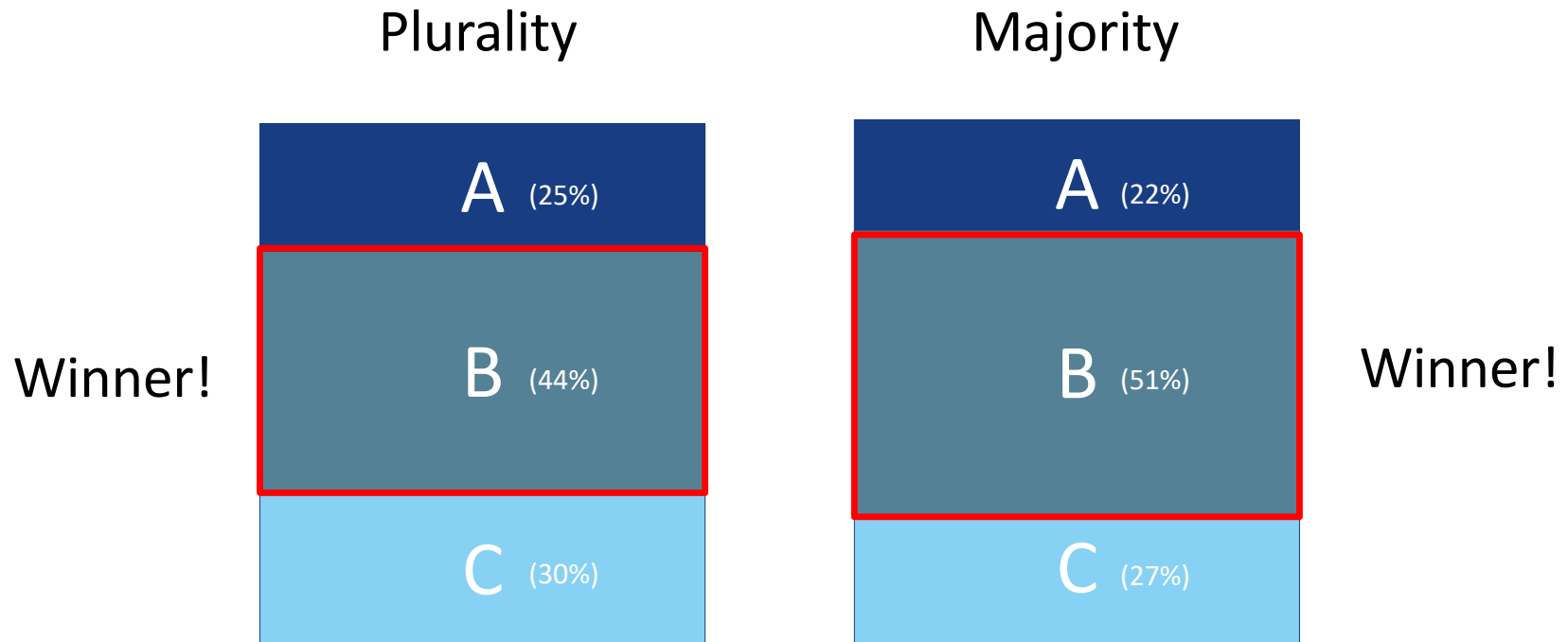
Outcome:

A = 2	
B = 3	→ Winner
C = 1	→ 'Lost vote'

PLURALITY VS MAJORITY



PLURALITY VS MAJORITY



MAJORITY RUNOFF ELECTION

Highest two go into *runoff election*

This second round of elections only has the highest two options

Requires losing agents to make a new choice among the two winner

Can lead to initial *tactical voting*

SEQUENTIAL MAJORITY VOTING

- Votes are given sequentially for 1 out of 2 options
- The winning options continues against the next option
- The last winning option is selected as overall winner

Each agent needs preferences among all options available – at least during voting

Sequence of votes can determine the winner *a priori*

An option is called *Condorcet Winner* if any sequence yields the same outcome

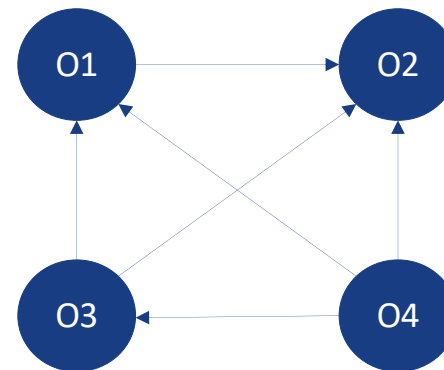
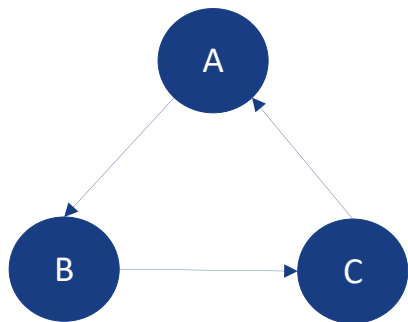
MAJORITY GRAPHS

Majority Graphs represent the preferences of *all* voters over *all* options

Also called *Tournament* if the majority graph is

- **complete:** for two options A and B; either A defeats B or *vice versa*
- **asymmetric:** if A defeats B, B cannot defeat A, and
- **irreflexive:** an option cannot defeat itself

Nodes represent an option, edges represent a majority preferring one option over the other

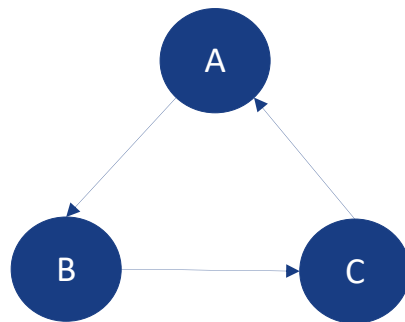


O4 is a Condorcet Winner

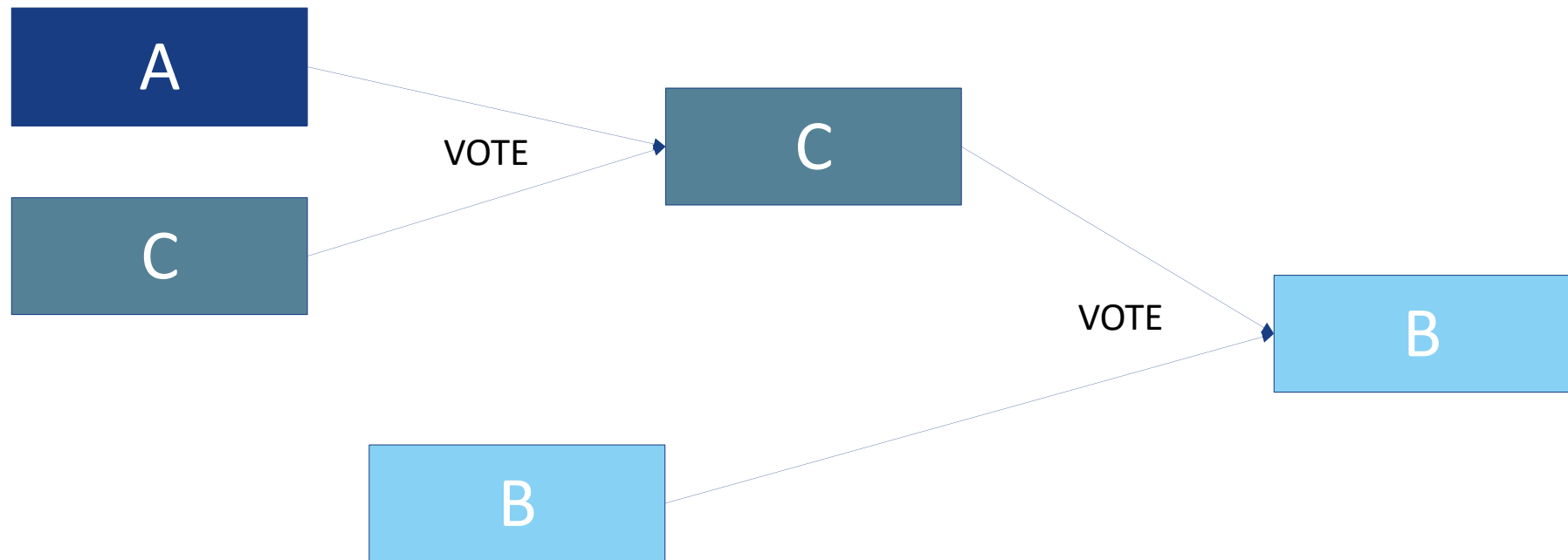
33 voters have preferences of the form $A \succ B \succ C$

33 voters have preferences of the form $C \succ A \succ B$

33 voters have preferences of the form $B \succ C \succ A$



SEQUENTIAL MAJORITY VOTING



BORDA COUNT

Agent votes by ranking all options

Points are given proportional to the rank

- Different approaches to distribute points are available

Points of all voters are added up and highest sum wins

Borda count is prone to tactical voting

How to deal with options not ranked? (=truncated ballots)

- Give 0 absolute points to unranked
- Consider only as many options as are ranked among all voters
- Give 0 relative points to unranked and adjust absolute points accordingly start from $1/3$, instead of $1/4$

Option	Rank	Relative points	Absolute points
O4	1	1.00	4
O2	2	0.75	3
O1	3	0.50	2
O3	4	0.25	1

Example for one agent with 4 options

Abs points = rel points * nr options

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How to deal with options not ranked? (=truncated ballots)

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- Consider only as many options as are ranked among all voters
- Give 0 relative points to unranked and adjust absolute points accordingly (more points for ranked options when selecting less of the available options)

Option	Rank	Relative points	Absolute points
O4	1	1.00	4
O2	2	0.75	3
O1	3	0.50	2
O3	4	0.25	1

Example for one agent with 4 options

Abs points = rel points * nr options

The votes for where to hold the conference were:

	51	25	10	14
1st choice	Seattle	Tacoma	Puyallup	Olympia
2nd choice	Tacoma	Puyallup	Tacoma	Tacoma
3rd choice	Olympia	Olympia	Olympia	Puyallup
4th choice	Puyallup	Seattle	Seattle	Seattle

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	51	25	10	14
1st choice 4 points	Seattle $4 \cdot 51 = 204$	Tacoma $4 \cdot 25 = 100$	Puyallup $4 \cdot 10 = 40$	Olympia $4 \cdot 14 = 56$
2nd choice 3 points	Tacoma $3 \cdot 51 = 153$	Puyallup $3 \cdot 25 = 75$	Tacoma $3 \cdot 10 = 30$	Tacoma $3 \cdot 14 = 42$
3rd choice 2 points	Olympia $2 \cdot 51 = 102$	Olympia $2 \cdot 25 = 50$	Olympia $2 \cdot 10 = 20$	Puyallup $2 \cdot 14 = 28$
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3rd choice 2 points				
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Adding up the points:

Seattle: $204 + 25 + 10 + 14 = 253$ points

Tacoma: $153 + 100 + 30 + 42 = 325$ points

Puyallup: $51 + 75 + 40 + 28 = 194$ points

Olympia: $102 + 50 + 20 + 56 = 228$ points

Under the Borda Count method, Tacoma is the winner of this vote.

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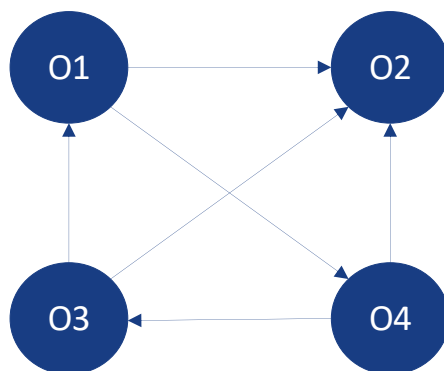
SLATER RANKING

Minimise the disagreement between majority graph and social ranking/social choice

Social choice is a theoretical concept with preference aggregation of all agents at its core

For each possible preference order measure the **degree of disagreement** with majority graph

Degree of disagreement = edges that need to be 'flipped' (NP-hard to compute)



For $O1 > O2 > O4 > O3$ as social choice, disagreement is 3
($O3, O1$), ($O3, O2$), ($O4, O2$) would need to be flipped

For $O4 > O3 > O1 > O2$ as social choice, disagreement is 1
($O4, O1$) is only edge that needs to be flipped

TACTICAL VOTING AND GERRYMANDERING

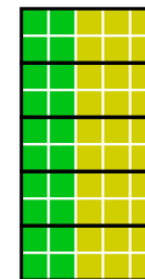
Tactical voting: voting for the candidate with the highest chances of winning instead of the actual favoured candidate

- In Plurality votes: Lets assume *A* and *C* are more similar than *B* and *C*. Voters for *C* might *a priori* decide to vote for *A* to not 'waste their votes'
- In Borda counts: Rank *A* higher even than *C* even though *C* might be our *sincere choice*

Gerrymandering: when votes are location dependent – changing the shape of the location can change the outcome of the vote.

- Often utilised in political votes (e.g. US)

50 Precincts
60% Yellow
40% Green



5 DISTRICTS
5 Yellow
0 Green



5 DISTRICTS
3 Yellow
2 Green



5 DISTRICTS
3 Green
2 Yellow

AUCTIONS

Efficient way to assign resources among competing agents

Single commodity

- English auctions
- Dutch auctions
- First-price sealed-bid auctions
- Vickrey auctions

Bidders have private/true valuation of commodity. Bidders make *profit* by paying less than their true valuation.

ENGLISH AUCTION

Best known type of auctions, often used for art and antiquity auctions.

1. Auctioneer presents an initial *reservation price* for the commodity – this can be 0. This is also considered the initial bid.
2. Agents present their own bids, exceeding the currently highest bid.
3. If there are no more bids, the agent with the highest bid wins the auction.



Without time limit, agents can approach their true valuation from a lower starting-point, allowing to pay less than the true valuation.

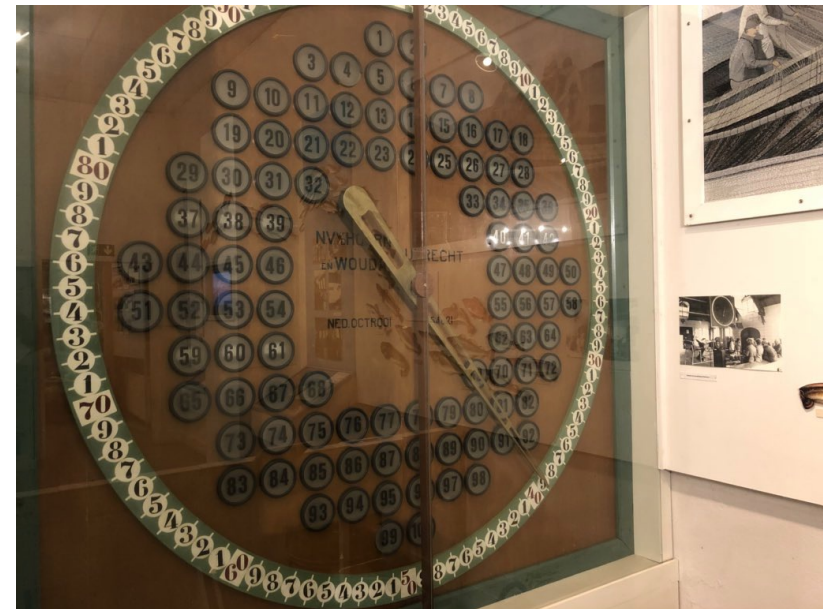
Simple dominant strategy: keep bidding as long as the current cost is below your valuation.

DUTCH AUCTIONS

Originally used in The Netherlands to sell estates, but also flowers, vegetables, and fruits.

Requires only a single bid.

1. Auctioneer starts with an artificially high bid
2. Auctioneer lowers price periodically – originally done using a ‘clock’.
3. First bid acquires commodity to price at bid.



Strategy for an agent is to bid below value as otherwise there is no profit. This requires speculation that others do not bid at higher price.

FIRST-PRICE SEALED-BID AUCTIONS

Also called *single-shot auctions*.

Can be held asynchronously.

1. Bidder present single bid to the auctioneer.

Bidder do not see the bids of others but only the auctioneer knows the highest bid.

2. Winner pays price of highest bid

No subsequent rounds of bidding

Strategy is equal to Dutch Auctions – to bid below own true valuation but higher than the second highest bid.
The goal is to bid as little more as possible as the second highest bid.

There is no general solution on how to determine this.

VICKREY AUCTION

Most unusual and counterintuitive auction.

Also called *second-price sealed-bid*.

1. Similar to first-price sealed-bid, bidders submit bids to auctioneer without others knowing the value of the bid.
2. **Highest bid wins** auction
3. Bidder pays only **price of second** highest bid.

VICKREY AUCTIONS

Dominant strategy is bidding with true valuation (truth revealing)

- Bidding lower only beneficial when this results in profit
- Bidding lower might result in not getting the commodity
- Bidding higher might result in paying over the true valuation

Payoff is $v - b_2$ if you win, and 0 if you lose.

$b > v \rightarrow b > b_2 > v$, in which case you will win the auction with a negative surplus $v - b_2$

$b < v \rightarrow v > b_2 > b$, 0 payoff, but you could have won with $b = v$

BARGAINING

Bargaining or negotiation used to agree about task or resource allocation and division.

- ***Negotiation set*** represents the set of possible proposals an agent can make
- ***Protocol*** defines what proposals an agent can choose from the negotiation set
- ***Strategies*** determine what proposals an agent will make.
- ***Rules*** determine what proposals an agent will accept and reject.

BARGAINING

Different approaches

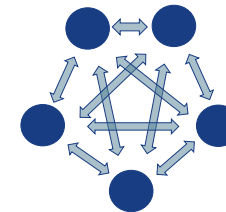
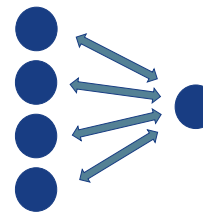
- **One-to-one:** simplest bargaining setting
- **Many-to-one:** one can choose from many proposals (e.g. auctions)
- **Many-to-many:** with n agents, this can be considered as

$n(n - 1)/2$ parallel one-to-one negotiations



One-to-one negotiation

Many-to-one negotiation



Many-to-many negotiation

PATIENT AND IMPATIENT AGENTS

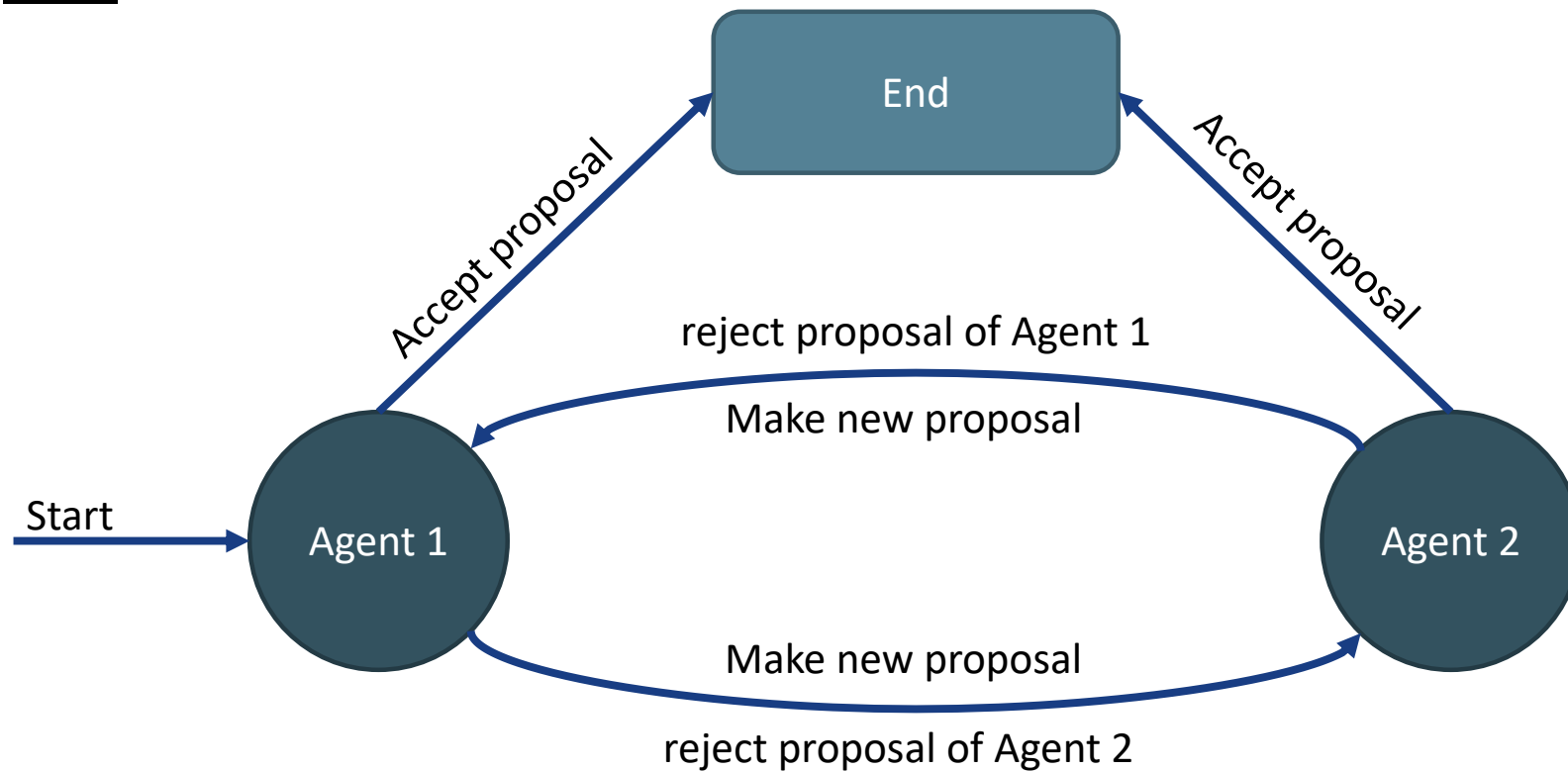
Patient agents **have no incentive** to change or accept different offer

- They will reject any other offer and not change their own
- Agents might get stuck in an impasse

Impatient agents are **required to strike a deal**

- They will try to adjust their offer or might accept the other agent's proposal
- Agents need to adjust their valuation of their offer over time

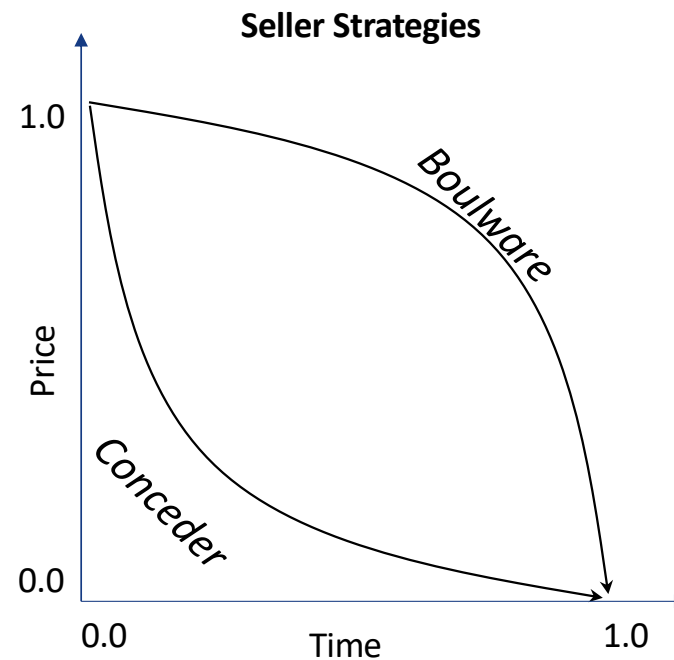
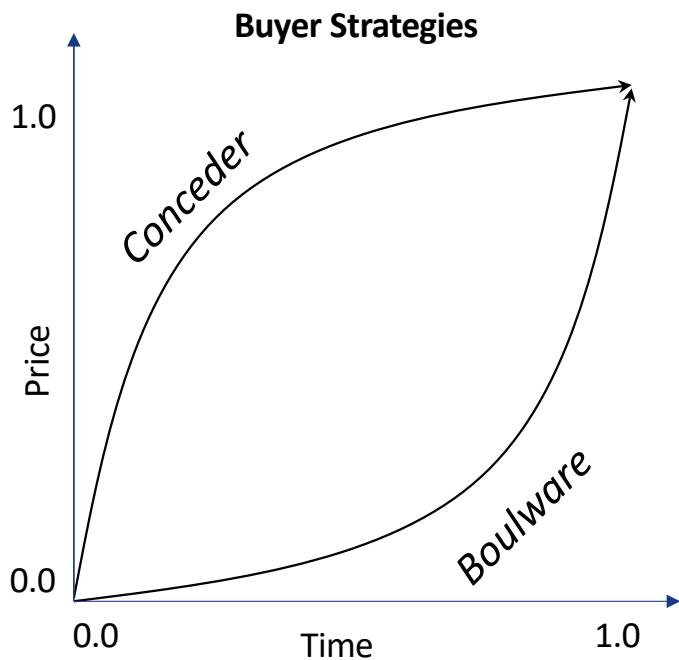
ALTERNATING OFFERS PROTOCOL



CLASSICAL STRATEGIES

Conceder: quickly lower price (seller) / pay higher price (buyer) at first but slower later on

Boulware: hold back at first but lower price (seller) / pay more (buyer) later on



15 MINUTES QUESTIONS

Questions group 1:

Explain the exploitation vs exploration trade-off in RL.

How does Ant Colony Optimization work?

Questions group 2:

Discuss different applications where we can use emergent approaches.

Compare sarsa to q-learning.

Questions group 3:

Compare different reasoning models for agents.

Describe different nature-inspired approaches.

EXERCISE

Use any of the methods mentioned to have a group of 5 agents to decide which one gets to do a particular task.

- Locations of agents and task are known
- Agents are within communication range
- You can make other assumptions if needed.